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Supplementary Methods

Analysis of existing datasets

In this report we reanalyzed four published articles that include both raw scores of psychiatric inventories and data from a cognitive task with confidence rating (Hoven, Luigjes, et al. 1; Katyal et al. 2; Rouault et al. 3; Seow & Gillan 4). Two datasets, Rouault et al. (3) and Seow and Gillan (4), used the exact original inventories from the factor analysis by Gillan et al. (5). Importantly, both reported an association between heightened mean confidence and CIT, as well as lowered mean confidence with AD. Hoven, Luigjes, et al. (1), included all questionnaires used by Gillan et al. (5), deviating only by using a different anxiety measure. Notably, Hoven, Luigjes, et al. (1) also reported an association between heightened mean confidence and CIT. The fourth and most recent dataset, Katyal et al. (2) (Experiment 2), used a shortened and optimized version of the Gillan et al. (5) transdiagnostic battery (6), comprising 71 items drawn from the same questionnaires used in the other datasets. Crucially for our analyses, this dataset also included two infrequency items, the gold-standard attention checks (7), enabling analyses focused on item-level sensitivity to inattentive responding (Analysis 4).

1. Rouault et al. (3), in which participants performed a perceptual discrimination task (decide which of two boxes has more dots in it) and rated their subjective confidence on a 6-point scale after each perceptual decision. We focused on experiment 2, which included the full pool of psychiatric questionnaires and shared the original analysis.

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2. Seow and Gillan (4), in which participants performed a predictive inference task (position a bucket to catch a flying particle) and rated their subjective confidence on a 100-point scale after making each prediction.

3. Hoven, Luigjes, et al. (1), in which participants performed the same perceptual discrimination task as in Rouault et al. (3) and rated their subjective confidence on a probabilistic 50-point scale after each decision.

4. Katyal et al. (2), in which participants completed a gamified task framed as helping imaginary residents of "Fruitville" collect fruit. The task interleaved perception trials, requiring a perceptual discrimination (judging whether there were more berries than raspberries), and memory trials, in which participants were briefly shown fruits and then asked to identify which fruit had been presented. Participants rated their subjective confidence on a continues slider ranging from 0 to 100%.

The following nine questionnaires were administered in both Rouault et al. (3) and Seow and Gillan (4) studies to assess various psychiatric symptoms: the Alcohol Use Disorder Identification Test (AUDIT) to measure alcohol addiction (8), the Apathy Evaluation Scale (AES) to assess apathy (9), the Self-Rating Depression Scale (SDS) to evaluate depression (10), the Eating Attitudes Test (EAT-26) for eating disorders (11), the Barratt Impulsivity Scale (BIS-11) to measure impulsivity (12), the Obsessive-Compulsive Inventory – Revised (OCI-R) to assess obsessive-compulsive disorder (13), the State-Trait Anxiety Inventory (STAI) for trait anxiety (14), the Short Scales for Measuring Schizotypy (SSMS) to assess schizotypy (15), and the Liebowitz Social Anxiety Scale (LSAS) for social anxiety (16). In Hoven, Luigjes, et al. (1)

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the same set of questionnaires was administered, except for the anxiety measure, where the Generalized Anxiety Disorder 7-item scale (GAD-7; 17, was used instead of the State-Trait Anxiety Inventory (STAI; 14). In Katyal et al. (2), an optimized item set developed by Hopkins et al. (6) was used, comprising 71 items drawn exclusively from the SDS, STAI, OCI, LSAS, BIS, EAT, and AES questionnaires, to derive transdiagnostic dimension scores.

In this analysis, we excluded the SSMS questionnaire because its binary scoring renders the measure of skewness irrelevant.

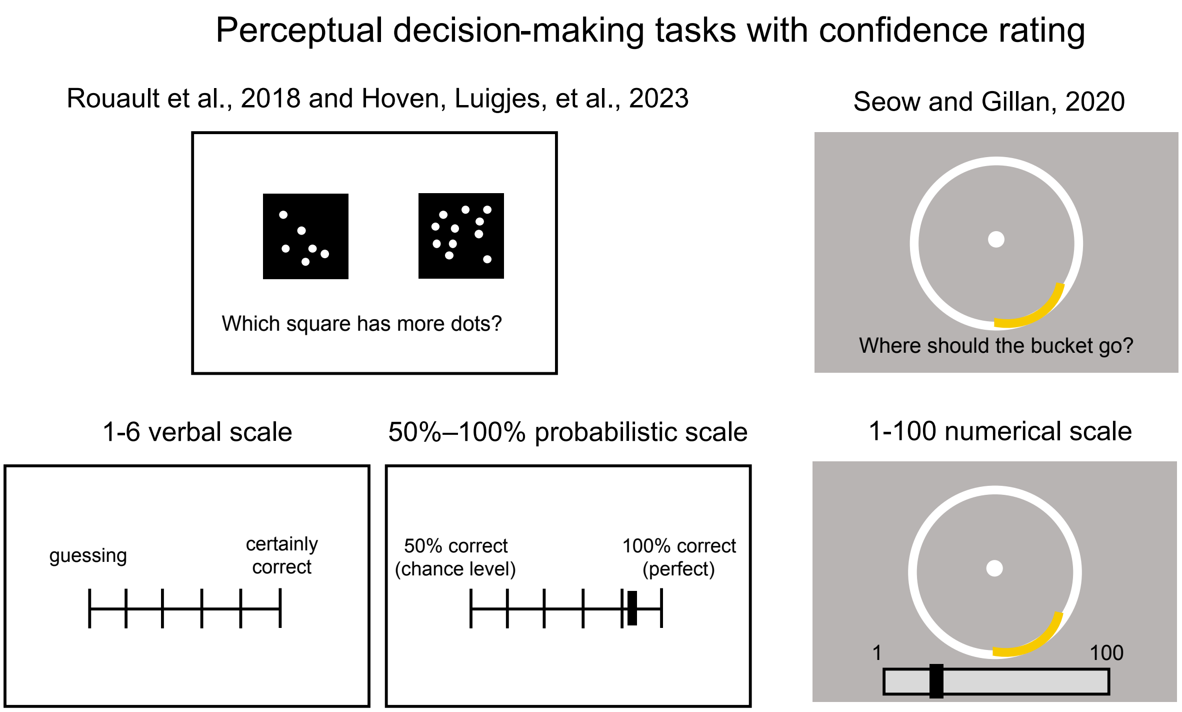

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Figure A1 *Illustration of Perceptual Decision-Making Tasks Re-analyzed in This Report.* Left: in the dot comparison task used by Rouault et al. (2018) and Hoven, Luigjes, et al. (2023), participants decided which of two squares contains more dots and then rated their decision confidence on a 6-point verbal scale (Rouault et al., 2018) or 50-point probabilistic scale (Hoven, Luigjes, et al., 2023). Right: in the predictive inference task used by Seow and Gillan (2020), participants positioned a bucket (yellow arc on the circle edge) to catch a flying particle and then rated their confidence in that they would catch the particle on a 100-point sliding scale.

Experiment

Perceptual decision-making task

The number of dots in each square varied across trials, with the difference between the two squares adjusted by a staircase procedure. One square contained a fixed number of 313 dots, while the other square had either more or fewer dots, depending on the trial's difficulty level. With a difference of fewer dots creating a more difficult task. Each square contained 625 possible positions for black dots (arranged in a 25x25 grid). The specific positions of the dots within each square were randomly selected from these 625 possible locations on every trial, and the square with the greater number of dots (the target) was randomly assigned to appear on the left or right side of the screen on each trial.

The experiment was programmed in jsPsych (18) and the experiment code and a demo of the task is available at https://github.com/self-model/BIRDAM. The order of experimental events was determined pseudo-randomly by the Mersenne Twister pseudorandom number generator, initialized to ensure registration time-locking (19).

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Comprehension questions

Participants answered the following two comprehension questions:

1. "If you are certain you made the correct judgment, where on the scale would you place your confidence from 50% 'Guessing' to 100% 'Certainly correct'?"
2. "If you are completely unsure whether you made a correct judgment, where on the scale would you place your confidence from 50% 'Guessing' to 100% 'Certainly correct'?"

Infrequency items

Each psychiatric questionnaire included two "infrequency" items to assess inattentive responding. Specifically, we used the following four items, (the first written by us and the last two adapted from Zorowitz et al. (20):

1. I was worried about the leprechauns who guard the hidden treasure (expected answer: '0- not at all').
2. I often rearrange the furniture in my home to prepare for the arrival of magical beans (expected answer: '0- not at all').
3. I find that relying on food and water is essential to my survival (expected answer: '4- Most of the time'; '3- Good part of the time').
4. I am worried about the canine World Cup (expected answer: '1- A little of the time').

Content-neutral items

We used 14 neutral items (a mixture of items adapted from Greenleaf (21) and ones created by us) to assess participants' global tendencies in using the rating scale (ranging from: 1- "Strongly agree" to 5- "Strongly disagree"). These items were intended to be heterogeneous in content, therefore, not expect to share common content, and neutral in the sense that, as a group, they are not minimally related to psychopathology.

1. I think quantitative information is difficult to understand.
2. When I go shopping, I find myself spending very little time checking out new products and brands.
3. Everyone should use a mouthwash to help control bad breath.
4. A college education is very important for success in today's world.
5. I like to visit places that are totally different from my home.
6. I work very hard most of the time.
7. I will probably have more money to spend next year than I have now.
8. I think fashion is irrelevant.
9. I believe there are relatively few different breeds of cats.
10. I think the moon is very far from earth.
11. As I see it, Madrid is a small place.
12. Book covers are important in my opinion.
13. These days, matchboxes are no longer useful.
14. I find the taste of apples different to that of pears.

Acquiescence correction procedure

For the acquiescence correction procedure, we first calculated each participant's mean response to the 14 neutral items, which served as the acquiescence proxy. We then used this score to control for acquiescence. For example, for the OCI-R correction procedure we performed the following steps:

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1. Fitted a linear model predicting each item rating from the participant's mean neutral rating score: OCIR_item_rating ~ mean_neutral_rating

2. Took the residuals for each item, i.e., the part of the rating not explained by acquiescence.

3. Summed the residuals across items.

4. Rescaled the total score to the original OCI-R scale.

The resulting index represents the OCI-R total score corrected for acquiescence and scaled to the same range as the original scores. We performed the same correction procedure for the SDS questionnaire as well.

Staircase Procedure

A staircase procedure was used to adjust the task difficulty based on participants' performance. The difference in the number of dots between the two squares (task difficulty) was initially set to 40 and then adjusted according to participants' accuracy: following a 2-down 1-up procedure with a step size of 2 and a minimum difference of 0. At the limit, this procedure converges to a proportion of 72% correct responses.

Reproducibility

We used R [Version 4.3.2; (22)] and the R-packages cowplot [Version 1.1.3; (23)], effectsize [Version 0.8.7; (24)], ggpubr [Version 0.6.0; (25)], groundhog [Version 3.1.2; (26)], lme4 [Version 1.1.35.3; (27)], lmerTest [Version 1.3.3; (28)], moments [Version 0.14.1; (29)], papaja [Version 0.1.2; (30)], patchwork [Version 1.2.0; (31)], polycor [Version 0.8.1; (32)], psych [Version 2.4.3; (33)], and tidyverse [Version 2.0.0; (34)] for all analyses.

Supplementary analysis

Linear mixed model analysis of item reversal effects on item-confidence correlation

In order to control for the possible effect of questionnaire content on the correlation between standard and reversed items with confidence, we performed a linear mixed model predicting item-correlation with confidence from item coding (standard/reversed) with random intercept for questionnaire (correlation ~ reversed + (1|questionnaire)). Results revealed that in both datasets reversed items showed lower correlations with confidence compared to standard items, even when accounting for questionnaire-level variation in the model: Seow and Gillan (4) ($\hat{\beta} = -0.17$, 95% CI $[-0.20, -0.15]$, $t(84.24) = -13.45$, $p < .001$), Rouault et al. (3) ($\hat{\beta} = -0.11$, 95% CI $[-0.14, -0.09]$, $t(85.97) = -8.92$, $p < .001$), Hoven, Luigjes, et al. (1) ($\hat{\beta} = -0.10$, 95% CI $[-0.12, -0.07]$, $t(65.01) = -6.73$, $p < .001$). We also tested a model allowing the effect of reversal to vary across questionnaires (random slopes), but model diagnostics indicated this model was too complex for the available data.

Linear mixed model analysis of skewness effects on item-confidence correlation

To examine whether the relationship between item skewness and item-confidence correlation was consistent across different questionnaires, we fitted a linear mixed models

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predicting item-confidence correlation from item-level skewness, including random intercepts and random slopes for questionnaire (conf_cor ~ skewness + (1 + skewness | questionnaire)). Results revealed a significant positive effect of skewness in all three datasets, even when allowing the strength of the relationship to vary across questionnaires: Rouault et al. (3) ($\hat{\beta} = 0.03$, 95% CI $[0.01, 0.05]$, $t(5.66) = 2.65$, $p = .040$), Seow and Gillan (4) ($\hat{\beta} = 0.12$, 95% CI $[0.08, 0.17]$, $t(6.02) = 6.00$, $p < .001$) and Hoven, Luigjes, et al. (1) ($\hat{\beta} = 0.03$, 95% CI $[0.01, 0.05]$, $t(4.87) = 3.47$, $p = .019$). The Hoven, Luigjes, et al. (1) model showed signs of overfitting, likely due to the smaller number of anxiety items (7 items in the GAD-7 vs. 20 items in the STAI used in the Rouault et al. (3) and Seow and Gillan (4) datasets). Therefore, we fitted an additional model with random intercept only (conf_cor ~ skewness + (1 | questionnaire)). This model also revealed a significant positive effect of skewness on item–confidence correlation ($\hat{\beta} = 0.01$, 95% CI $[0.00, 0.02]$, $t(150.74) = 2.10$, $p = .038$), supporting a general within-questionnaire relationship that is not solely attributable to between-questionnaire differences.

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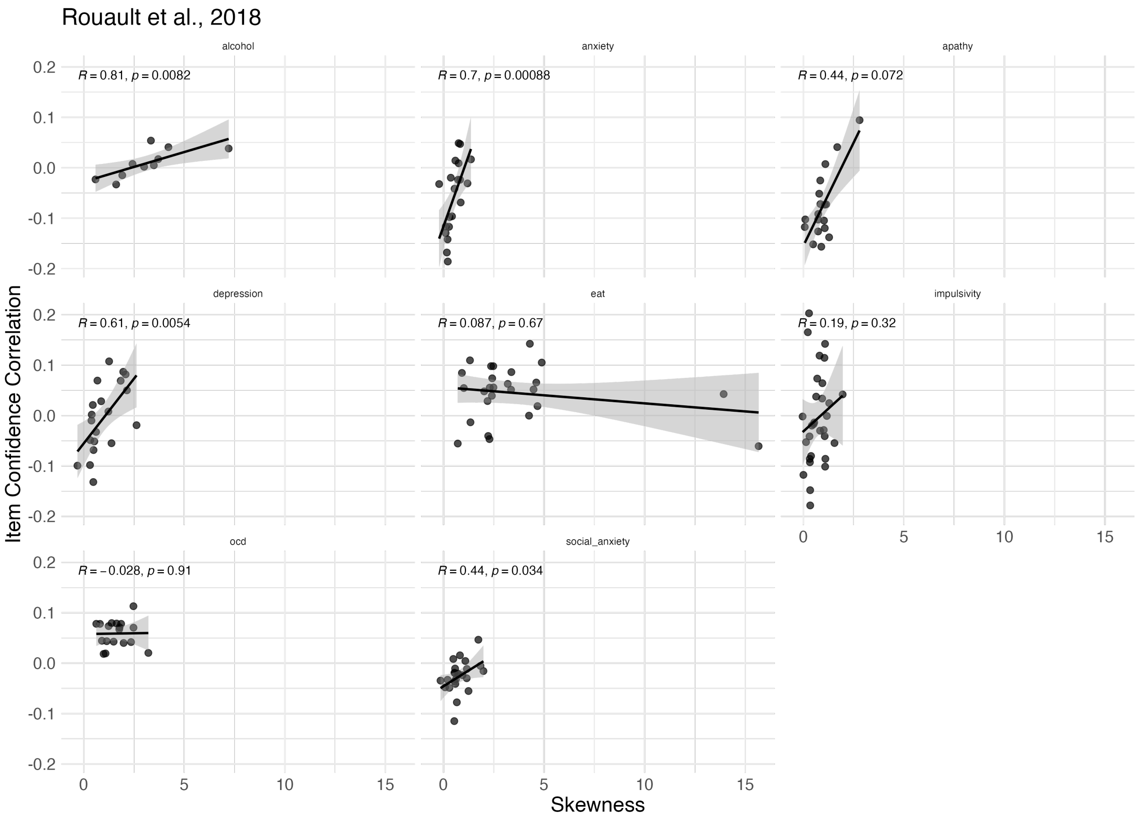

Figure A2 *Within-Questionnaire Analysis of Item Skewness and Item Confidence Correlations in Rouault et al. (2018).* Each panel represents a different questionnaire, and each point represents an individual item. The x-axis shows the skewness of item responses. Y-axis shows the correlation between item ratings and confidence across participants. The black line represents the linear fit with 95% confidence intervals (gray shading). Spearman correlation coefficients (R) and corresponding p-values are shown for each questionnaire.

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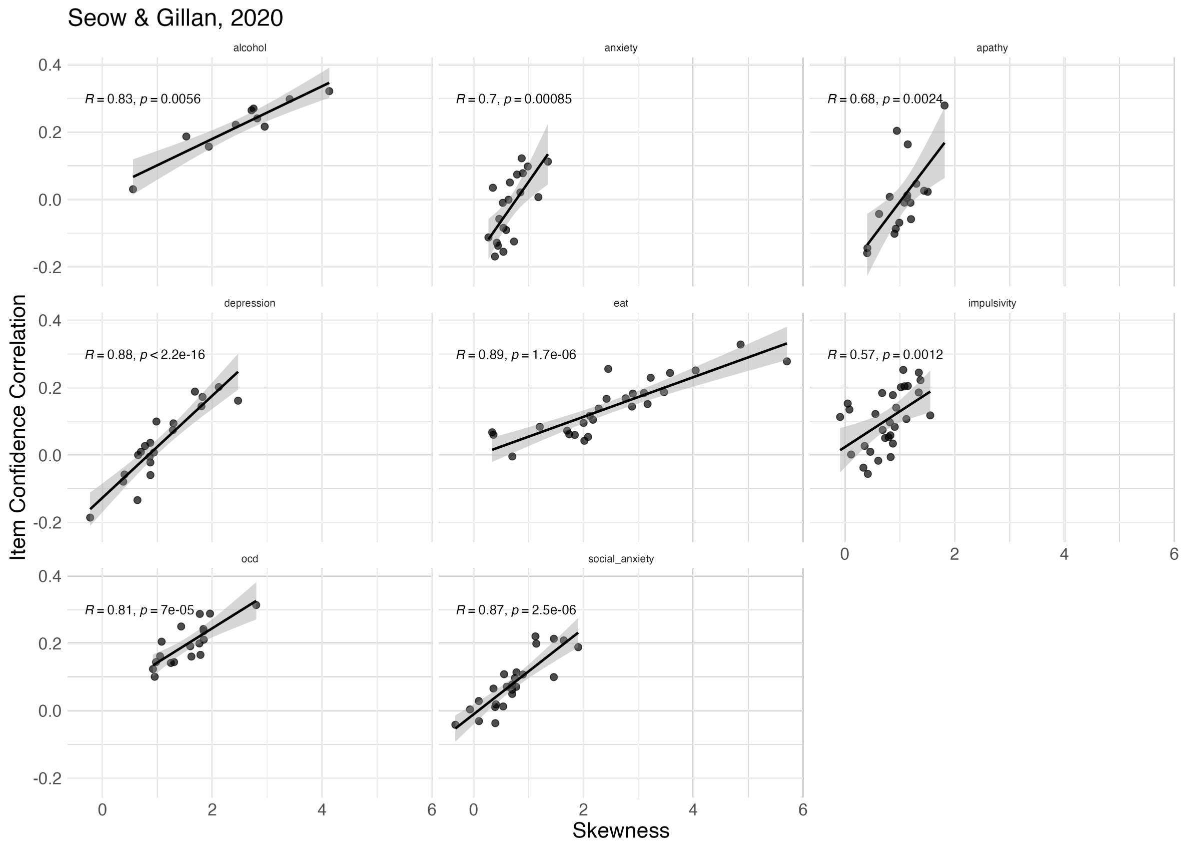

Figure A3 *Within-Questionnaire Analysis of Item Skewness and Item-Confidence Correlations in Seow and Gillan (2020). Same conventions as Figure A2.*

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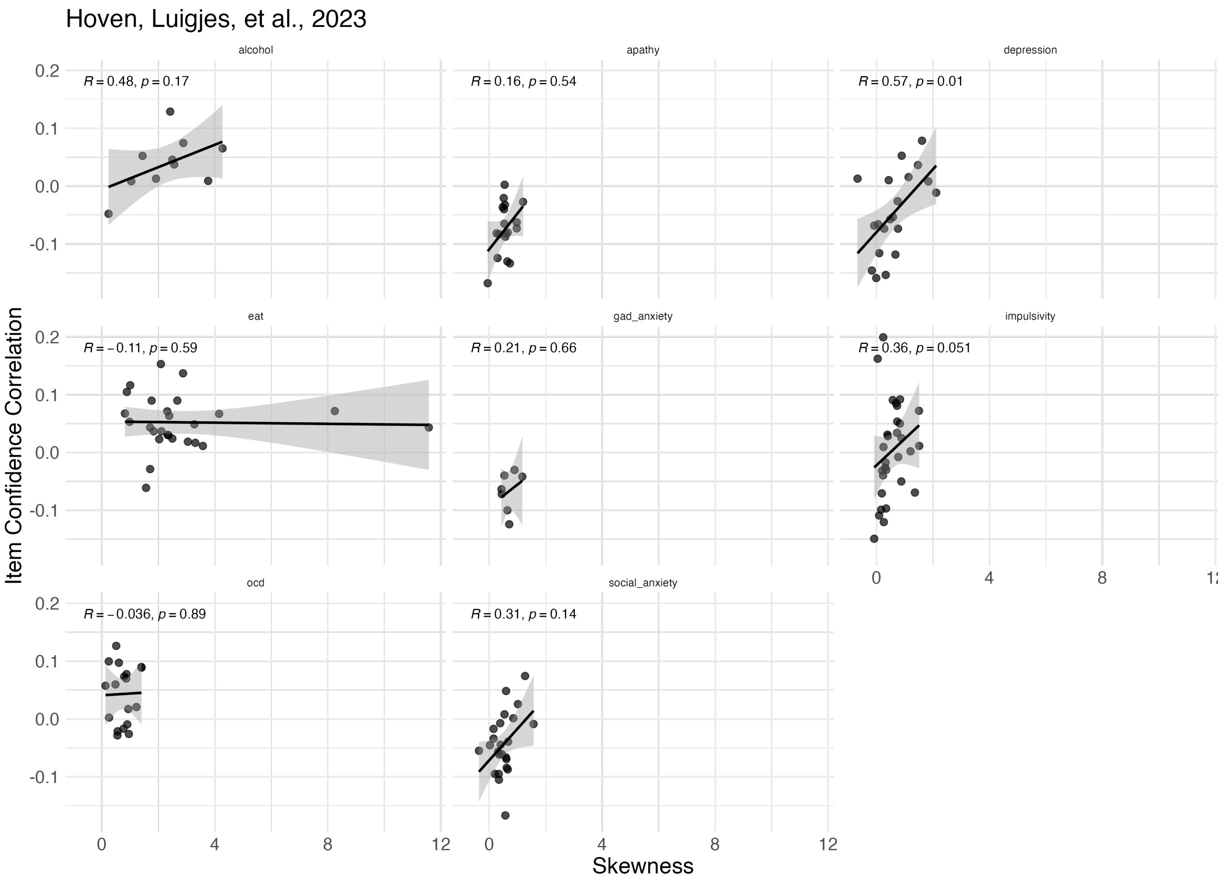

Figure A4 *Within-Questionnaire Relationship of Item Skewness and Item-Confidence Correlations in Hoven, Luigjes, et al. (2023).* Same conventions as Figure A2.

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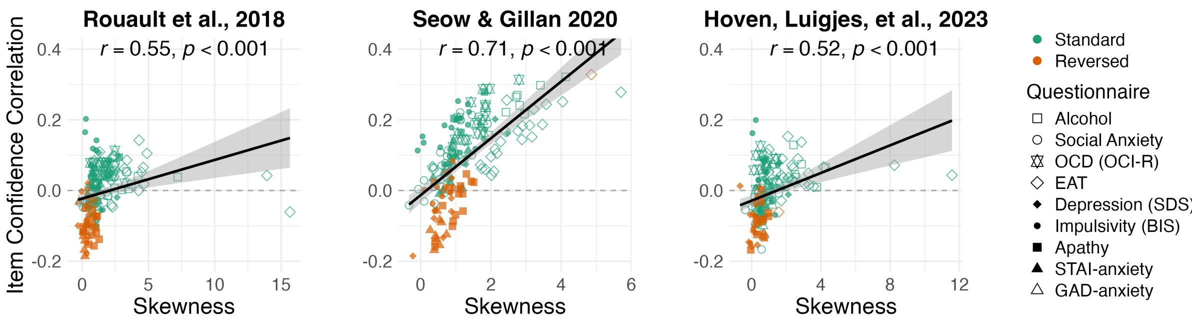

Figure A5 *Correlation Between Item-Level Skewness and Confidence with Extreme Skewed Items Included.* Same convention as Figure 3.

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***Figure A6** The Effect of Reversed Items and Skewness on the CIT and AD Dimensions with Extreme Skewed Items Included. Same conventions as Figure 4.*

Associations between transdiagnostic dimensional weights with skewness and coding direction in Hoven, Luigjes, et al. (2023)

As Hoven, Luigjes, et al. (1) used the GAD-7 instead of the STAI to measure anxiety (as was used in the original factor analysis performed by Gillan et al. 5), we extracted factor weights from Gillan et al. (5), only for the items that were common across the two studies. Which were all the other measures except the GAD-7. Consistent with Analysis 3 in both Rouault et al. (3) and

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Seow and Gillan (4), our reanalysis of Hoven, Luigjes (1) revealed a positive correlation between item weight and item skewness on the CIT dimension, such that more skewed items contributed more strongly to the CIT factor ($r_s = .56$, $p < .001$; Figure A7, left panel). Similarly, the AD dimension showed a pattern consistent with the other two datasets, with reversed items of low skewness loading heavily on the AD factor (Figure A7, right panel).

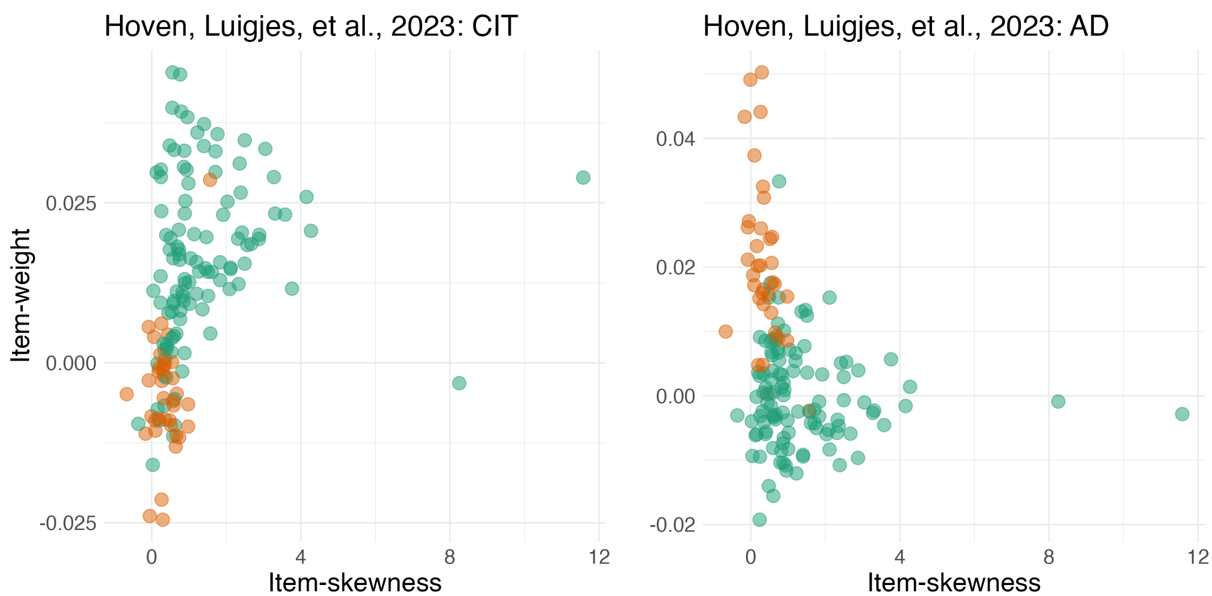

Figure A7 *The Effect of Reversed Items and Skewness on the CIT and AD Dimensions in Hoven, Luigjes, et al. (2023).* Same conventions as Figure 4.

Linear mixed model analysis of skewness effects on CIT item weights correlation

To examine whether the relationship between item skewness and CIT item weight was consistent across different questionnaires, we fitted linear mixed-effects models predicting CIT item weight from item-level skewness. In the Rouault et al. (3) and Seow and Gillan (4) datasets, we used models with random intercepts and random slopes for questionnaire (CIT ~ skewness + (1 + skewness | questionnaire)). Both models revealed a significant positive effect of skewness on CIT item weight: Rouault et al. (3) ($\hat{\beta} = 0.0085$, 95% CI $[0.0033,\ 0.0137]$, $t(6.13) = 3.20,\ p = .018$), Seow & Gillan (4) ($\hat{\beta} = 0.0101$, 95% CI $[0.0051,\ 0.0151]$, $t(5.24) = 3.97,\ p = .010$). In the Hoven, Luigjes, et al. (1) dataset, the model with random slopes did not converge, likely due to the smaller number of anxiety items (7 items in the GAD-7 vs. 20 items in the STAI used in the Rouault et al. (3) and Seow and Gillan (4) datasets). We therefore fitted a model with random intercepts only (CIT ~ skewness + (1 | questionnaire)). This model also showed a significant positive effect of skewness ($\hat{\beta} = 0.0014$, 95% CI $[0.0000,\ 0.0027]$, $t(141.48) = 1.99,\ p = .048$). Together, these models suggests that the relationship between skewness and CIT item weight is not dependent on any specific questionnaire and is observed on average within questionnaires.

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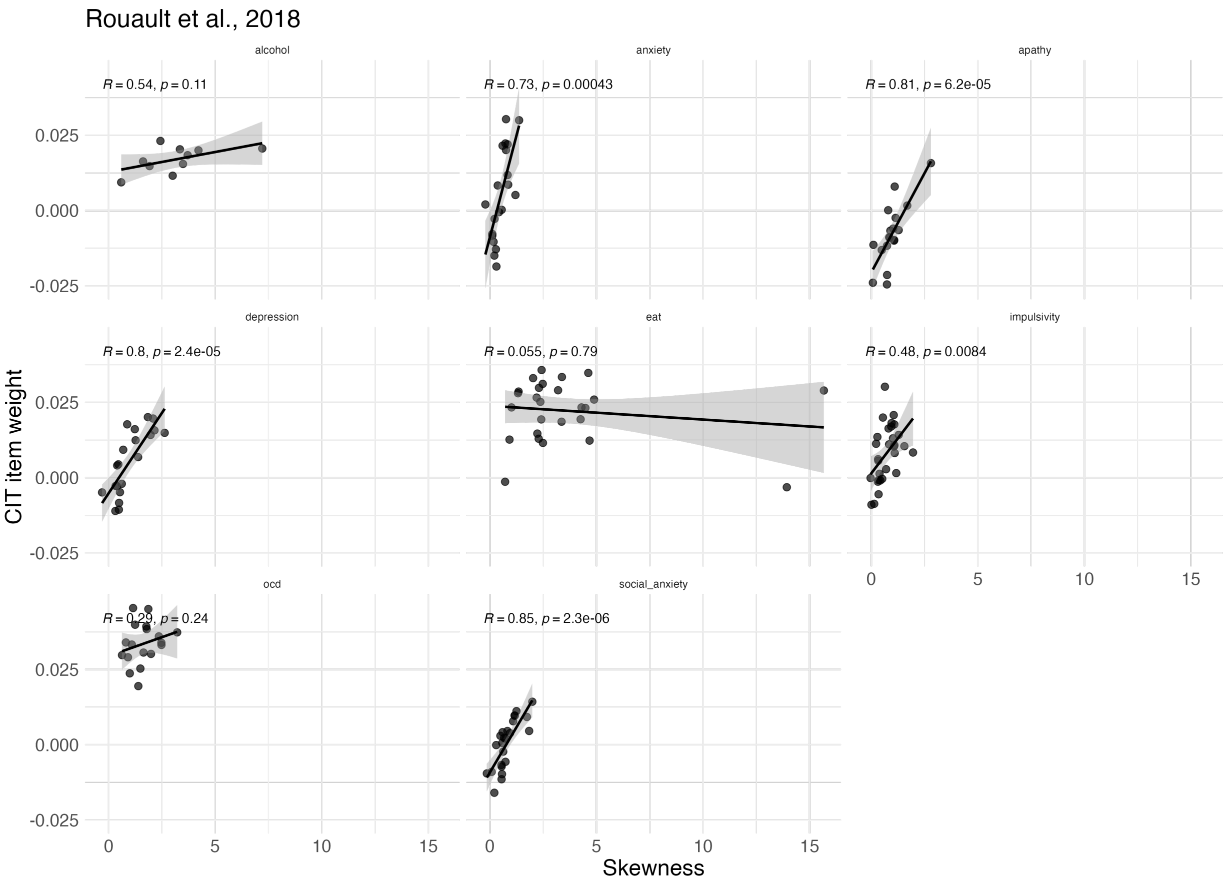

Figure A8 *Within-Questionnaire Analysis of Item Skewness and CIT Item Weights in Rouault et al. (2018).* Each panel represents a different questionnaire, and each point corresponds to an individual item. The x-axis shows the skewness of item responses; the y-axis shows the corresponding CIT item weight. The black line represents the linear fit with 95% confidence intervals (gray shading). Spearman correlation coefficients (ρ) and p-values are displayed within each panel.

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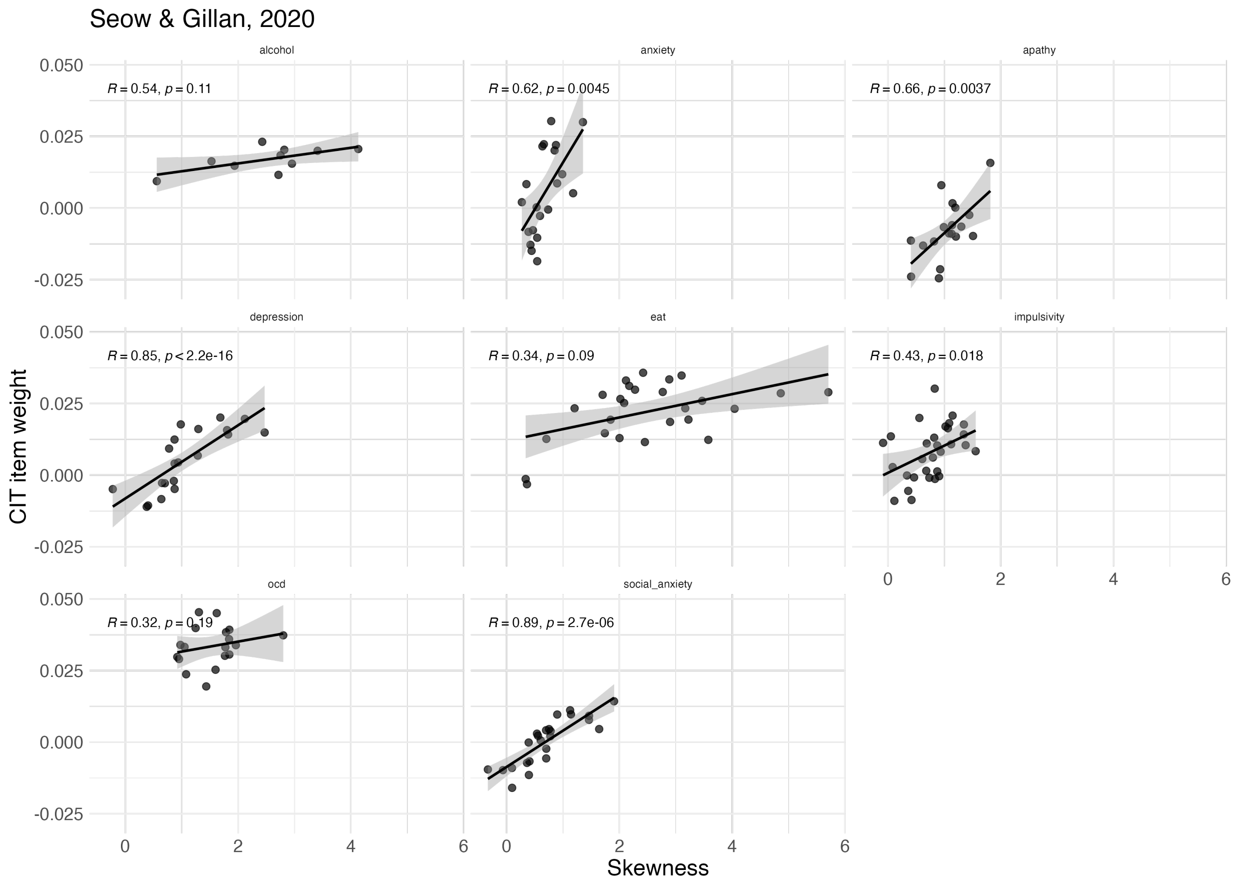

Figure A9 *Within-Questionnaire Analysis of Item Skewness and CIT Item Weights in Seow and Gillan (2020)*. Same conventions as Figure A8.

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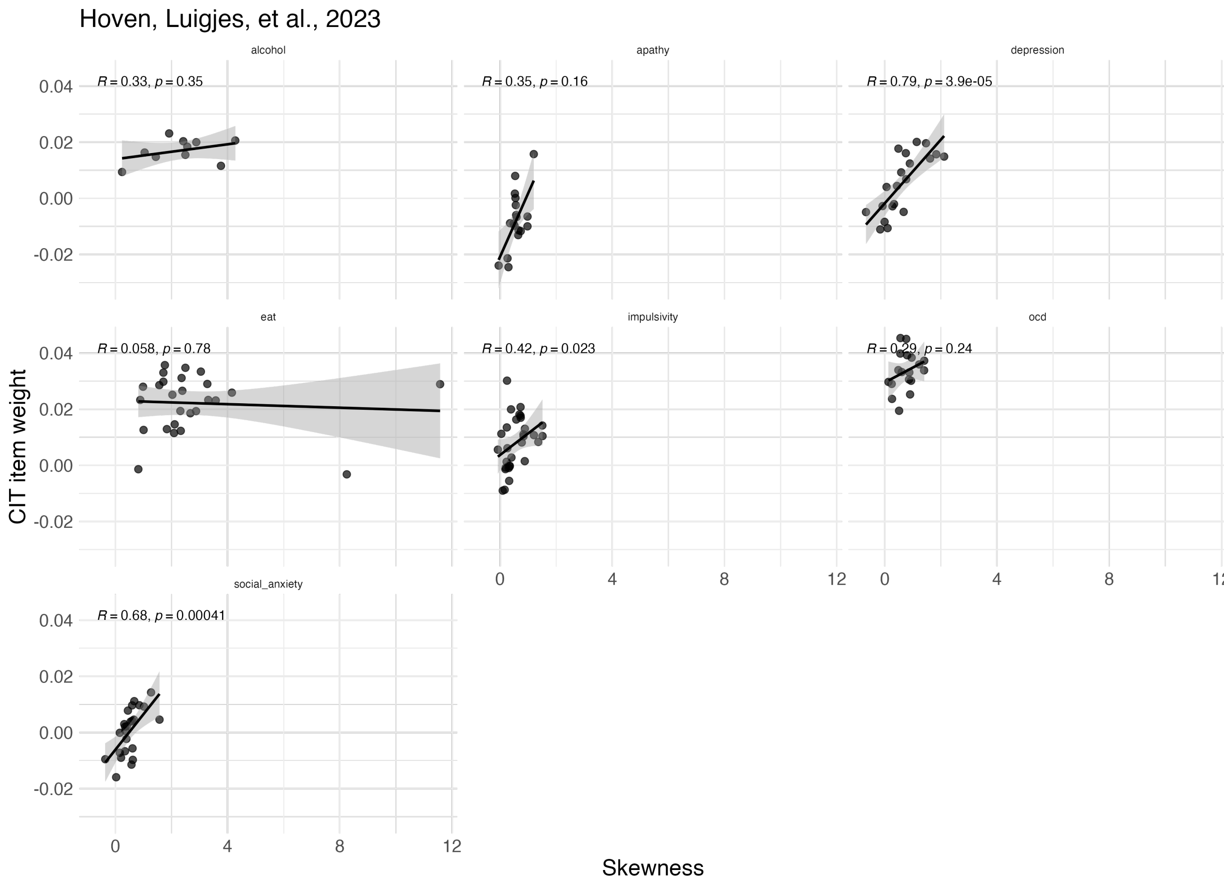

Figure A10 *Within-Questionnaire Analysis of Item Skewness and CIT Item Weights in Hoven, Luigjes, et al. (2023*). Same conventions as Figure A8.

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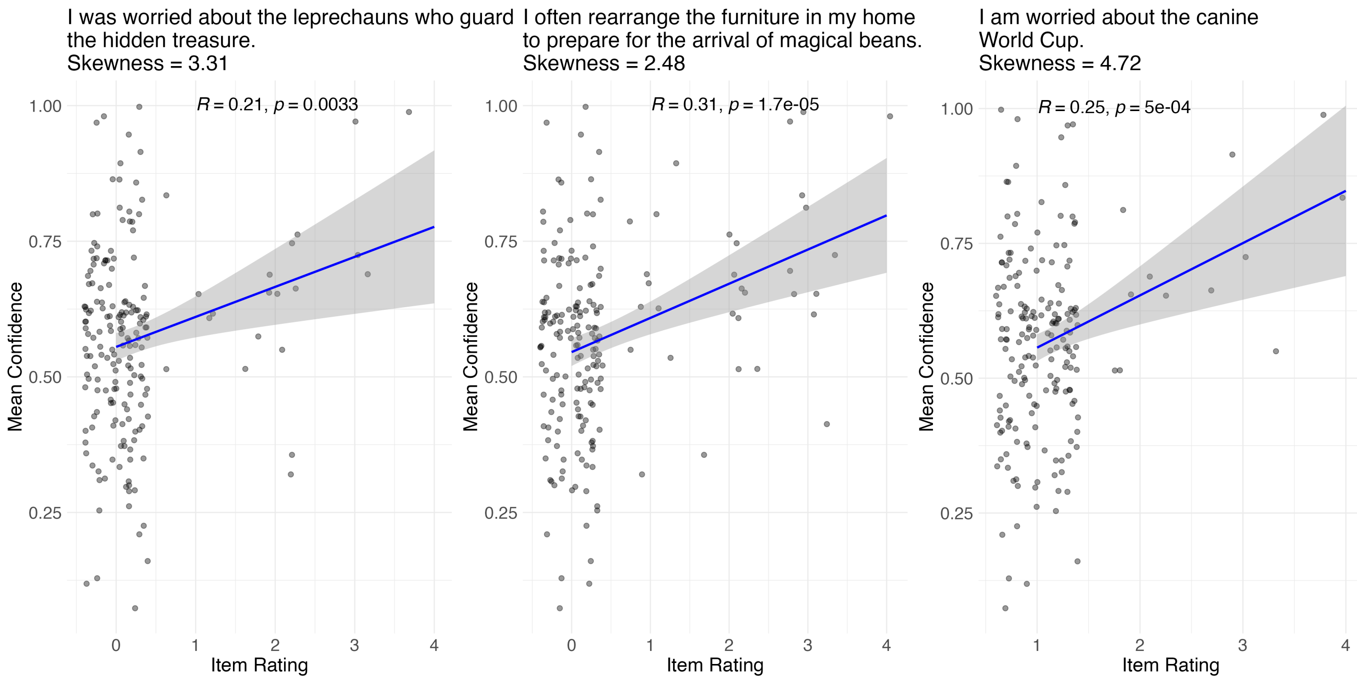

Figure A11 *Highly Skewed Infrequency Items Show Positive Correlation Between Item Endorsement and Mean Confidence.* Each panel displays the relationship between endorsement of a highly skewed infrequency item (x-axis) and mean decision confidence (y-axis). The blue line represents a linear fit with 95% confidence intervals (gray shading). Pearson correlation coefficients (*R*) and *p*-values are reported in each plot. The first two items were administered within the OCI-R questionnaire: *"*I was worried about the leprechauns who guard the hidden treasure*"* and *"*I often rearrange the furniture in my home to prepare for the arrival of magical beans.*"* The third item, *"*I am worried about the canine World Cup,*"* was embedded in the SDS questionnaire.

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Effects of inattentiveness and acquiescence on confidence controlling for sex and age

We conducted multiple regression analyses to examine the effects of inattentiveness and acquiescence on mean confidence while controlling for sex and age. In the first model with inattentiveness, sex, and age as independent variables, we found that inattentiveness significantly predicted mean confidence even when controlling for age and sex ($b = 0.09$, 95% CI [0.03, 0.14], $t(184) = 3.03$, $p = .003$). In the second model with acquiescence, sex, and age as independent variables, we found that acquiescence also significantly predicted mean confidence when controlling for sex and age ($b = 0.13$, 95% CI [0.06, 0.20], $t(184) = 3.75$, $p < .001$).

Sex differences in inattentive responding

In our sample, the distribution of Sex (Male/Female self-reported on Prolific; two participants who did not report their sex were excluded from this analysis) differed between attentive and inattentive participants. Among attentive responders, there was no significant sex difference (Male: n = 69; Female: n = 73), whereas among inattentive participants, males were more frequent (Male: n = 30; Female: n = 16). This effect was only marginally significant in a chi-square test ($\chi^2(1, n=188) = 3.21,\ p = .073$)

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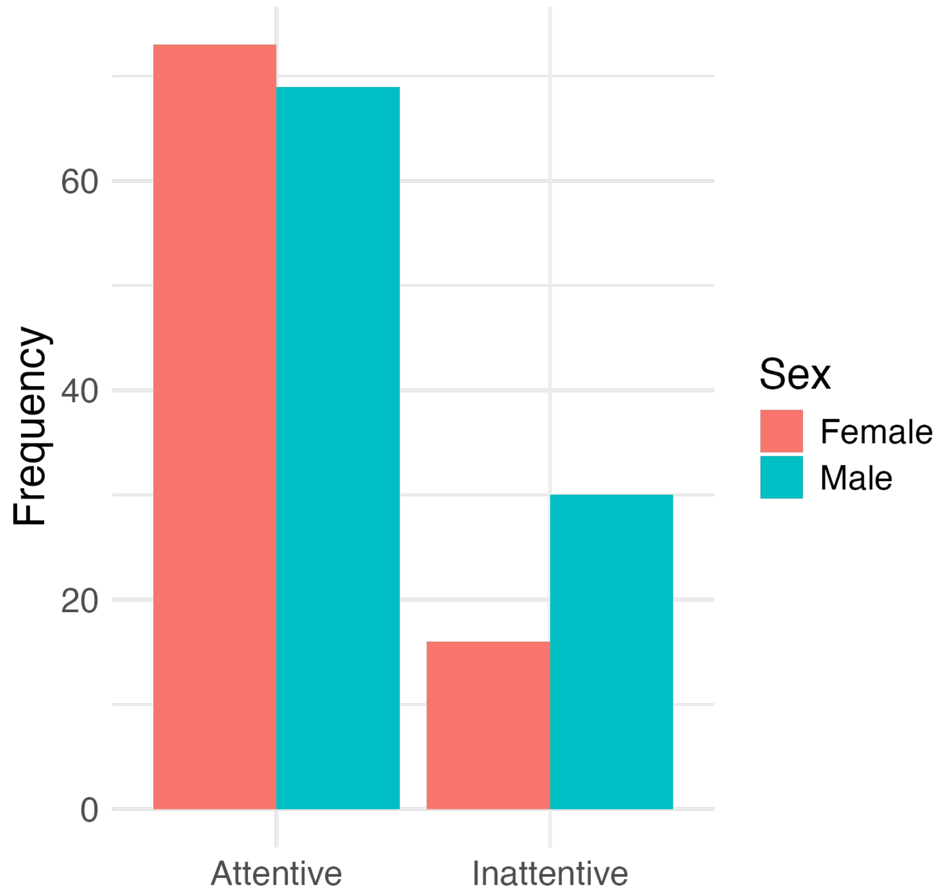

Figure A12 *Sex differences in inattentiveness prevalence.*

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Figure A13 *Mean Confidence by Attentiveness and Sex.* Scatter plot showing mean confidence ratings across attentive and inattentive responders, separated by sex. Blue and pink points represent attentive and inattentive participants, respectively. Error bars indicate the standard error for each subgroup.

Inattentive responders encounter an easier task due to staircasing

As inattentive responders generally perform worse than attentive responders, they encounter an overall easier task difficulty when using a staircase procedure (see Supplementary Methods for details on this procedure). This occurs because more mistakes lead to a reduction in task difficulty. In our experiment, task difficulty was defined as the difference in dots (increment) between two stimuli. The greater the difference, the easier the task, as the distinction between the two squares becomes more apparent. Therefore, we computed difficulty as the negative of the increment:

$$difficulty = -\,increment$$

Higher increments result in easier perceptual decisions. The mean difficulty was then calculated for each participant to assess overall task difficulty. We found that inattentive participants encountered significantly easier task on average compared to attentive participants, $t(188) = 5.20,\ p < .001$ (figure A14 left panel). We also found a negative correlation between task difficulty and mean confidence ratings, indicating that as the task became easier, mean confidence increased, $p = .004$ (figure A14, right panel).

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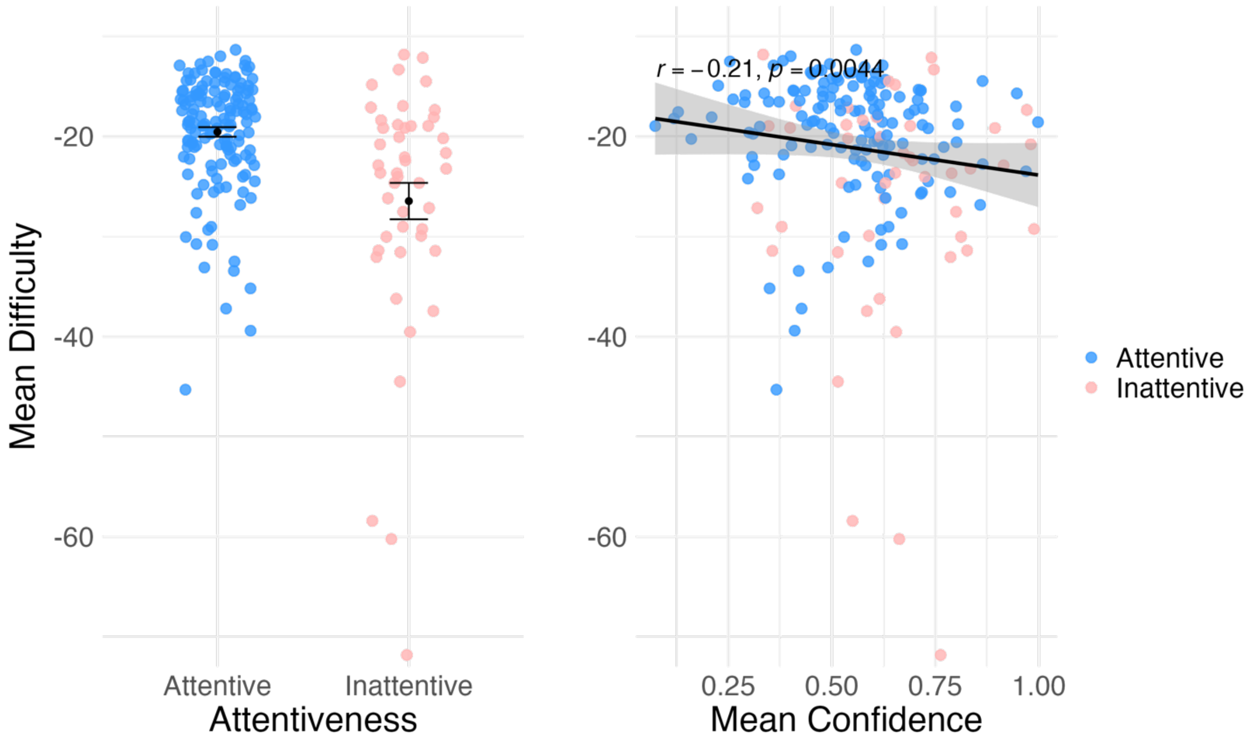

Figure A14 *Task Difficulty and Inattentiveness.* Left panel: mean difficulty ratings as a function of attentiveness, with data points for attentive and inattentive responses shown in red and blue, respectively. Black markers and error bars represent the mean and standard error, respectively. Right panel: mean confidence as a function of mean difficulty, with attentive and inattentive responses again differentiated by color. A negative correlation is observed, as shown by the black regression line (shaded area represents the confidence interval).

Analysis excluding participants that failed the confidence comprehension check

At the end of the perceptual task, participants' comprehension was assessed with two comprehension items (see Supplementary Methods). Our pre -registered plan was not to exclude participants according to these items. However, as an exploratory analysis, we report our main analyses after excluding participants that failed the comprehension items. Initially, we only included participants who gave confidence ratings above 85 in response to the item: “If you are certain you made the correct judgment, where on the scale would you place your confidence from 50% ‘Guessing’ to 100% ‘Certainly correct’?” and below 65 in response to the item: “If you are completely unsure whether you made a correct judgment, where on the scale would you place your confidence from 50% ‘Guessing’ to 100% ‘Certainly correct’?”. Only 52 participant passed these stringent criteria. Instead, we adopted a more lenient criterion, and included all participants who gave higher rating to the first item than to the second item. This resulted in a total of 149 participants. We then tested our two main hypotheses which remained significant (hypothesis 1: $t(46.07)=4.53,\ p<.001$, hypothesis 2: $r_s=.31,\ p<.001$).

A bootstrap permutation test for response style effects on OCI-R-confidence correlations

To validate that the reduction in correlation between total OCI scores and mean confidence ratings was stronger than expected by chance, we implemented a bootstrap permutation test. The procedure consisted of the following steps:

For each of 1,000 iterations, we:

1. Removed a random subset of 46 participants (equivalent to the number of inattentive participants in our original analysis) from the total sample.
2. Randomly assigned mean rating values (from content neutral items) to each participant, sampling without replacement from the original distribution of mean ratings.

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3. For each OCI-R item we fitted a linear regression model predicting item response from the randomly assigned mean rating. Then, we extracted the residuals to use as an acquiescence-controlled item score.

4. Calculated new total OCI-R scores for each participant by summing the residualised item scores and scaling the corrected totals to match the original scale range (0-62). (Scaling was done for visualization purposes, and, being a linear transformation, does not affect the correlation with confidence).

5. Computed the Pearson correlation between corrected OCI-R total scores and mean confidence ratings. This procedure maintained the same sample size reduction and mathematical correction process as our main analysis but broke the relationship between participants and their mean ratings scores.

6. Computed the correlation drop between the original OCI-R total score confidence correlation coefficient and the correlation coefficient produced in each iteration, giving us an estimate of the expected drop in correlation.

The resulting distribution of correlation coefficients served as a null distribution, representing the expected drop in correlation coefficient if inattentive responding and acquiescence were not truly associated with OCI-R scores. We compared our observed drop in the correlation coefficient (r=0.17), calculated as the difference between the original correlation coefficient (r=0.28) and correlation coefficient observed after controlling for acquiescence and inattentiveness using actual mean ratings and removing truly inattentive participants ($r=0.11$), to this null distribution. We found that all 1,000 iterations produced smaller Pearson correlation coefficients drops than our observed value ($p < .001$), which suggests that the reduction in correlation in our main analysis represents a genuine effect of controlling for inattentiveness and acquiescence (Figure 5 panel J). We ran the same analysis for the SDS questionnaire. Out of 1,000 iterations in our null distribution, 626 produced correlation coefficients below our observed value, indicating that the observed reduction in correlation was not significantly different from what would be expected by chance ($p = .374$, Figure 5 panel K).

Model estimation of inattentive responding

We estimated the prevalence of inattentive responding through a detection rate analysis and model fitting procedure. First, a detection rate analysis was conducted to examine the effectiveness of using multiple infrequency items. For this analysis, we calculated the percentage of participants identified as inattentive based on combinations of one to four infrequency items. For each possible combination of k items (where $k = 1, 2, 3$, or 4), we computed the percentage of participants who failed at least one item within that combination. We then averaged these percentages across all possible combinations of the same size to obtain the mean detection rate for each number of items. We found the following inattentive responders detection rates: 11.18% with one item, 17.54% with two items, 21.45% with three items, and 24.21% with four items.

To estimate the total proportion of inattentive respondents in our sample, including those not detected by our infrequency items, we fit a model assuming that: (1) a fixed proportion of participants (x) respond inattentively, and (2) each inattentive participant has a probability (p) of being captured by any single infrequency item. Under this model, the probability of detecting an inattentive participant with n items is given by:

$$\text{Prob(detection)} = x \cdot \left(1 - (1 - p)^n\right)$$

Model parameters were estimated using a grid approximation approach. We created a 101 × 101 grid of possible parameter values, with both x (the proportion of inattentive participants) and p (the probability of an inattentive participants being captured by a single infrequency item) ranging from 0 to 1 in increments of 0.01. For each parameter combination, we calculated the negative log-likelihood (NLL) of observing our detection rates given those parameters:

$$\text{NLL}(x, p) = -\sum \log\left(\text{Likelihood}\left(\text{observed}_n / N, x, p\right)\right)$$

where the likelihood was calculated using the binomial probability mass function, with N representing the total number of participants. The parameter combination that minimized the NLL was selected as the best-fitting model.

NLL reached a minimum for x = 0.28 and p = 0.39, suggesting that 28% of participants were responding inattentively, with each inattentive participant having a 39% probability of being caught by any single infrequency item. The model's predictions closely matched the observed detection rates (Figure A15).

SI Appendix: Biased and Inattentive Responding Contributes to Apparent Metacognitive Biases in Mental Health

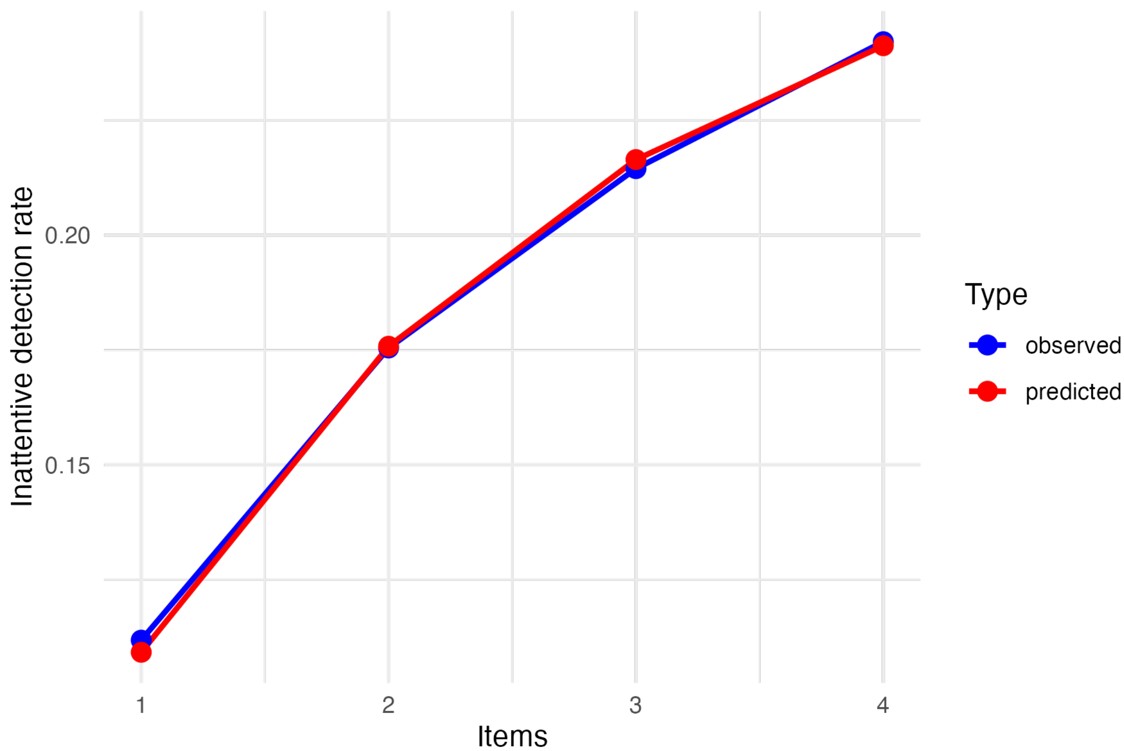

Figure A15 *Observed and Predicted Inattentive Detection Rates as a Function of the Number of Infrequency Items.* Observed detection rates (blue) represent the mean percentage of participants identified as inattentive across all possible combinations of k infrequency items. The red line represents the best-fitting model, which estimated that 28% of participants responded inattentively, with each inattentive participant having a 39% probability of being detected by a single infrequency item.

Simulation of the relationship between item skewness and factor weights

To assess the relationship between item skewness and factor weights, we conducted a simulation using R. We simulated responses from 200 participants on a 20-item questionnaire measuring two distinct latent traits. The first 10 items measured the first latent trait, while the remaining 10 items measured the second latent trait. Data generation followed these steps: First, two underlying traits (a and b) were simulated for each participant from normal distributions (M = 0, SD = 0.5). Responses were then generated by adding random normal error (SD = 1) to the relevant trait value, using the mean of the latent variable. Then, the cumulative density function of the normal distribution *N(0,1)* was used to translate individual responses to quantiles, ranging from 0 to 1. Then, quantile scores were projected back to ratings, using distributions of different levels of skewness for different items. This was done by using the inverse cumulative density function of beta distributions with shape parameters *b=2* for all items, and a ranging from 2 to 11. Finally, scores were binned into seven bins of equal width.

Factor Analysis Procedure

For each simulated dataset, we followed the factors analysis procedure of Gillan et al. (5). We calculated a heterogeneous correlation matrix using polychoric correlations (via polycor::hetcor() with Maximum likelihood estimation) to account for the ordinal nature of the data. We then conducted a factor analysis using the psych package (specifically psych::fa()) with maximum likelihood estimation, a fixed two-factor solution, and oblimin rotation ensuring the mean weight per factor is positive. Item skewness was measured using moment package, and Spearman correlations between item skewness and the corresponding factor weights were calculated. We then took the averaged correlation across the factors.

We repeated this procedure 1000 times using different random seeds. On average, we observed a pattern of weak negative correlations between item skewness and item weight (

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$M = -0.04$, 95% CI $[-0.04, -0.03]$, $t(999) = -14.40$, $p < .001$), meaning that item weights were slightly closer to zero for more skewed items. This effect is much weaker, and in the opposite direction, to what we find for the CIT dimension in both datasets.

SI Appendix: Biased and Inattentive Responding Contributes to Apparent Metacognitive Biases in Mental Health

Full Survey Questions and Instructions

Welcome instructions

Welcome to the second part of the study. In this part, you will be asked to respond to several statements about everyday experiences. Thank you for your time and cooperation.

OCI-R instructions

The following statements refer to experiences that many people have in their everyday lives. Mark the number that best describes how much that experience has distressed or bothered you during the past month.

	Not at all 0	A little 1	Moderately 2	A lot 3	Very much 4
I have saved up so many things that they get in the way.	○	○	○	○	○
I check things more often than necessary.	○	○	○	○	○
I get upset if objects are not arranged properly.	○	○	○	○	○
I feel compelled to count while I am doing things.	○	○	○	○	○

I find it difficult to touch an object when I know it has been touched by strangers or certain people.	○	○	○	○	○
I find it difficult to control my own thoughts.	○	○	○	○	○
I collect things I don't need.	○	○	○	○	○
I repeatedly check doors, windows, drawers, etc.	○	○	○	○	○

I get upset if others change the way I have arranged things.	○	○	○	○	○
I feel I have to repeat certain numbers.	○	○	○	○	○
I sometimes have to wash or clean myself simply because I feel contaminated.	○	○	○	○	○
I am upset by unpleasant thoughts that come into my mind against my will.	○	○	○	○	○

I avoid throwing things away because I am afraid I might need them later.	○	○	○	○	○
I repeatedly check gas and water taps and light switches after turning them off.	○	○	○	○	○
I need things to be arranged in a particular way.	○	○	○	○	○
I feel that there are good and bad	○	○	○	○	○

numbers.					
I wash my hands more often and longer than necessary.	○	○	○	○	○
I frequently get nasty thoughts and have difficulty in getting rid of them.	○	○	○	○	○
I was worried about the leprechauns who guard the hidden treasure.	○	○	○	○	○
I often rearrange the furniture in	○	○	○	○	○

my home to prepare for the arrival of magical beans.	

SDS instructions

For each item below, please check the option which best describes how often you felt or behaved this way during the past several days:

	1. A Little Of The Time	2. Some Of The Time	3. Good Part Of The Time	4. Most Of The Time
I feel down-hearted and blue.	○	○	○	○
Morning is when I feel the best.	○	○	○	○
I have crying spells or feel like it.	○	○	○	○
I have trouble sleeping at night.	○	○	○	○
I eat as much as I used to.	○	○	○	○
I still enjoy sex.	○	○	○	○

I notice that I am losing weight.	○	○	○	○
I have trouble with constipation.	○	○	○	○
My heart beats faster than usual.	○	○	○	○
I get tired for no reason.	○	○	○	○
My mind is as clear as it used to be.	○	○	○	○
I find it easy to do the things I used to.	○	○	○	○

I am restless and can't keep still.	○	○	○	○
I feel hopeful about the future.	○	○	○	○
I am more irritable than usual.	○	○	○	○
I find it easy to make decisions.	○	○	○	○
I feel that I am useful and needed.	○	○	○	○
My life is pretty full.	○	○	○	○
I feel that others would be better off if I were	○	○	○	○

dead.				
I still enjoy the things I used to do.	○	○	○	○
I find that relying on food and water is essential to my survival.	○	○	○	○
I am worried about the canine World Cup.	○	○	○	○

Content-neutral items inventory instructions

This next set of questions is designed to measure your general attitudes and your approach to answering questions. Please read each of the following statements carefully and rate how much you agree or disagree using the scale provided. There are no right or wrong answers.

	1 - Strongly disagree	2 - Somewhat disagree	3 - Neither agree nor disagree	4 - Somewhat agree	5 - Strongly agree
I think quantitative information is difficult to understand.	○	○	○	○	○
When I go shopping, I find myself spending very little time checking out new products and brands.	○	○	○	○	○
Everyone should use a mouthwash to help control bad	○	○	○	○	○

breath.					
A college education is very important for success in today's world.	○	○	○	○	○
I like to visit places that are totally different from my home.	○	○	○	○	○
I work very hard most of the time.	○	○	○	○	○
I will probably have more money to spend next year than I	○	○	○	○	○

have now.					
I think fashion is irrelevant.	○	○	○	○	○
I believe there are relatively few different breeds of cats.	○	○	○	○	○
I think the moon is very far from earth.	○	○	○	○	○
As I see it, Madrid is a small place.	○	○	○	○	○
Book covers are important in my	○	○	○	○	○

opinion.					
These days, matchboxes are no longer useful.	○	○	○	○	○
I find the taste of apples different to that of pears.	○	○	○	○	○

End of survey participants comments

Before we finish, we would appreciate your feedback on your experience with this survey. Please share any thoughts or comments you have. Thank you very much for your participation.

Over- and under-control of covariates in compulsivity: A reply to Gillan et al. (2025)

In a reply to an earlier version of this paper (published as a preprint on March 17, 2025; https://osf.io/preprints/psyarxiv/fzwu8_v1) (35), Gillan et al. (36) criticize several aspects of our analysis approach. Specifically, they criticize the low statistical power of our experiment, our use of single questionnaires instead of their CIT and AD dimensions, over interpretation of post-hoc measures and the omission of covariates from our analysis. They further show that, in a new dataset of unpaid volunteers, a correlation between compulsivity and confidence persists even when excluding inattentive participants. We address each of these points in turn.

1. The inclusion of covariates can produce spurious associations

Gillan et al. (36) argue that our failure to find a positive correlation between the OCI-R and confidence is due to not including sex, gender, and SDS scores in our statistical model. Specifically, they write:

> *"To illustrate the importance of including these well-established confounds, we added these covariates in a re-analysis of their data and recovered the often-reported significant negative association between depression and confidence that Sarna et al. fail to observe. Importantly, we also find that removing inattentive responders does not in fact reduce the associations between confidence and either the depression or OCD symptoms, in contrast to what they report (Table 1), with both OCD symptoms (p=.001) and depression (p=.002) remaining significantly and bidirectionally linked to confidence. The association between confidence and OCD, when controlling for depressive scores, age and gender, also remains significant even after correcting for acquiescence (p=.009)."*

They show (Table A1) that the OCI-R coefficient is nonsignificant when excluding inattentive participants and controlling for acquiescence ($b = 0.02$, 95% CI $[-0.01, 0.05]$, $t(140) = 1.31$, $p = .193$), but becomes significant when including age, sex and SDS scores in the model ($b = 0.04$, 95% CI $[0.01, 0.07]$, $t(137) = 2.66$, $p = .009$).

		Full sample (N=188)		Attentive Only (N=142)		Attentive Only + Acq. Corr. (N=142)	
		Est. (SE)	**p-value**	**Est. (SE)**	**p-value**	**Est. (SE)**	**p-value**
Model 1	Depression	-0.02 (0.01)	0.077	-0.02 (0.01)	0.075	-0.02 (0.01)	0.089
Model 2	OCD	0.05 (0.01)	<.001***	0.03 (0.01)	0.069	0.02 (0.01)	0.193
Model 3	Depression	-0.04 (0.01)	0.002**	-0.05 (0.02)	0.002**	-0.04 (0.02)	0.006**
	OCD	0.06 (0.01)	<.001***	0.05 (0.02)	0.001**	0.04 (0.02)	0.009**
	Age	-0.02 (0.01)	0.195	-0.02 (0.01)	0.153	-0.02 (0.01)	0.126
	Sex	0.06 (0.02)	0.0097**	0.05 (0.03)	0.041*	0.06 (0.03)	0.042*

Table A1 - from Gillan et al. (2025). Original table note: Re-analysis of Sarna et al. (2025). Models 1 and 2 are used by Sarna et al. (2025). Model 3 includes covariates for co-occurring mental health symptoms and demographics (age and sex) variables. "Acq. Corr." refers to acquiescence correction of clinical scores using the mean neutral rating. Two participants without sex data are removed.

It is worth highlighting that the increase in predictive value of OCI-R is *solely* due to the inclusion of the SDS regressor, not the sex and age ones. When omitting SDS from the model the OCI-R coefficient remains unchanged ($b=0.02$, 95% CI $[-0.01, 0.05]$, $t(138)=1.36$, $p=.175$), but a model that includes only SDS as a covariate produces a significant OCI-R coefficient ($b=0.04$, 95% CI $[0.01, 0.08]$, $t(139)=2.74$, $p=.007$).

Why does the inclusion of SDS in the model have this effect on the OCI-R coefficient? One interpretation is that the negative correlation between depression and confidence masks a positive correlation between obsessive-compulsiveness and confidence. If obsessive-compulsive individuals tend to be more depressed than their peers (indeed, the correlation between SDS and OCI-R scores in our sample was $r=.54$, 95% CI $[.42, .65]$, $t(140)=7.67$, $p<.001$ when excluding inattentive responders and controlling for acquiescence), it is conceivable that the positive correlation between OCI-R and confidence is suppressed by a stronger negative effect of depression. But an alternative is that OCD and confidence are uncorrelated, but the strong correlation between the OCI-R and SDS regressors artificially creates opposite effects on confidence. This is sometimes referred to as “conditioning on a collider” (37-39).

Consider the following simulation in which individual subjects are described by four independently and identically distributed latent (unobserved) variables: latent general psychopathology ($psychopathology_{latent} \sim N(0,1)$), latent depression ($depression_{latent} \sim N(0,1)$), latent obsessive-compulsive tendencies ($OC_{latent} \sim N(0,1)$), and latent confidence ($confidence_{latent} \sim N(0,1)$). Measured (observed) variables are linear combinations of latent variables: measured confidence negatively weighs latent depression, but is uncorrelated with latent obsessive compulsive tendencies ($confidence_{measured} = confidence_{latent} - 0.2 \times depression_{latent}$), OCI-R (corrected for acquiescence, and excluding inattentive responders) is defined to be a linear combination of latent psychopathology and latent obsessive-compulsive tendencies ($OCIR_{observed} = psychopathology_{latent} + OC_{latent}$), and SDS (corrected for acquiescence, and excluding inattentive responders) is defined to be a linear combination of latent psychopathology and latent depression ($SDS_{observed} = psychopathology_{latent} + depression_{latent}$).

Analysing the data of 1000 simulated participants, a linear model that predicts measured confidence from SDS (similar to m1 in Gillan et al. 36) correctly infers that SDS and measured confidence are negatively related ($b=-0.12$, 95% CI $[-0.18, -0.05]$, $t(998)=-3.59$, $p<.001$; Table A2- Model 1). Similarly, a linear model that predicts confidence from OCI-R (similar to Gillan et al. 36; Table 1- Model 2) correctly infers that OCI-R and measured confidence are independent ($b=0.02$, 95% CI $[-0.05, 0.09]$, $t(998)=0.61$, $p=.543$; Table A2- Model 2). And yet, when including both in the same model (like in their Model 3), both SDS and OCI-R appear as if they contribute to measured confidence, with opposite effects (SDS: $b=-0.18$, 95% CI $[-0.26, -0.10]$, $t(997)=-4.61$, $p<.001$, OCI-R: $b=0.11$, 95% CI $[0.04, 0.19]$, $t(997)=2.94$, $p=.003$; Table A2- Model 3). To reiterarte: this model was designed such that OCI-R and measured confidence are independent, but controlling for SDS makes them appear correlated.

		Est. (SE)	p-value
Model 1	Depression	-0.12 (0.03)	<.001***
Model 2	OCD	0.02 (0.03)	0.543
Model 3	Depression	-0.18 (0.04)	<.001***
	OCD	0.11 (0.04)	0.003***

Table A2 - *Results of a simulation in which OCI-R and* measured *confidence are designed to be independent, yet they appear correlated in Model 3 due to a collider bias. Linear models predicting measured confidence from simulated OCI-R and SDS scores.*

A similar effect can be observed in our raw data, before correcting for acquiescence effects and excluding inattentive participants. Since the OCI-R questionnaire has no reversed items, it is highly correlated with acquiescence. So, a linear model that predicts our proxy of acquiescence (mean rating to content-neutral items) from OCI-R scores reveals a positive relationship ($b = 0.27$, 95% CI $[0.13, 0.41]$, $t(186) = 3.85$, $p < .001$). The inclusion of reversed items in the SDS makes SDS total scores independent of acquiescence, and indeed a linear model that predicts mean neutral ratings from SDS total scores reveals no relationship ($b = -0.04$, 95% CI $[-0.19, 0.10]$, $t(186) = -0.59$, $p = .556$). Critically, however, when including both in the same model, the positive relationship with the OCI-R slightly strengthens ($b = 0.34$, 95% CI $[0.19, 0.49]$, $t(185) = 4.48$, $p < .001$) and, more dramatically, SDS now appears to correlate with acquiescence ($b = -0.17$, 95% CI $[-0.32, -0.02]$, $t(185) = -2.30$, $p = .022$). In sum, while the inclusion of two highly correlated regressors does change the nature of the inferences, it is very difficult to tell whether this change reflect the uncovering of a true association or the emergence of a spurious one. For this reason, for the purpose of our paper, focusing on raw correlations seemed more appropriate.

2. Fully controlling for covariates is hard

A central claim made by Gillan et al. (36) in their reply is that even when inattentive responding is well-controlled, the bidirectional association between the AD and CIT dimensions and confidence persists. To support this, they show data from Fox et al. (40), which included multiple attention checks to identify inattentive responders, yet still observed a significant association between CIT and confidence with no significant differences between attentive and inattentive participants (Figure A16). They interpret this as evidence that the effect is not driven by inattention. However, whether this finding truly challenges the response bias interpretation depends on the extent to which such datasets can be considered fully controlled for inattention and acquiescence.

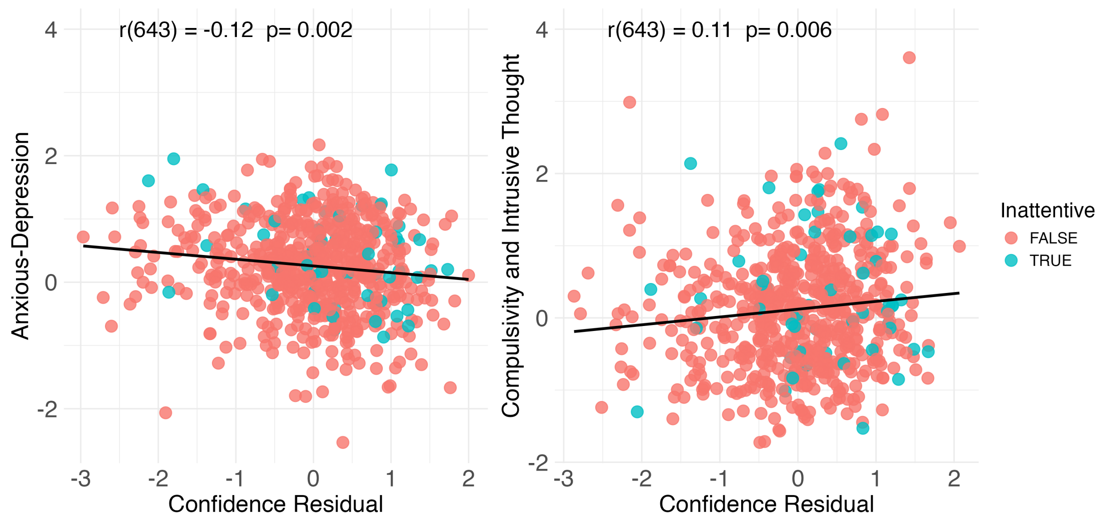

Figure A16 *- from Gillan et al. (2025).* Original figure note: Inattentive responding (in blue) has no impact on the associations between confidence and mental health dimensions in a large clinical sample from Fox et al. (2023). Confidence Residual refers to confidence scores residualised for age and gender and AD/CIT.

Consistent with inattention remaining only partly controlled, Fox et al. (40) reported only 8% of participants as inattentive, a rate much lower than the 24% detected in our data and below the inattention rates reported by Zorowitz et al. (20) (22% in general and in clinical samples). These lower rates likely reflect the way inattention was assessed—using only instructed-response items (e.g., “If you are paying attention to these questions, please select x”) rather than infrequency items, which are recommended as the gold standard (20; 41). Instructed-response items have been shown to be poor measure of inattention (42-43). As a result, Fox et al. (40) likely underestimated the true rate of inattentive responding in their sample, and the reported association between CIT and confidence is therefore still confounded by undetected inattention. This interpretation is consistent with the finding that inattentive participants in their data show the same pattern we report: higher levels of CIT, OCI-R, and confidence (Figure A17).

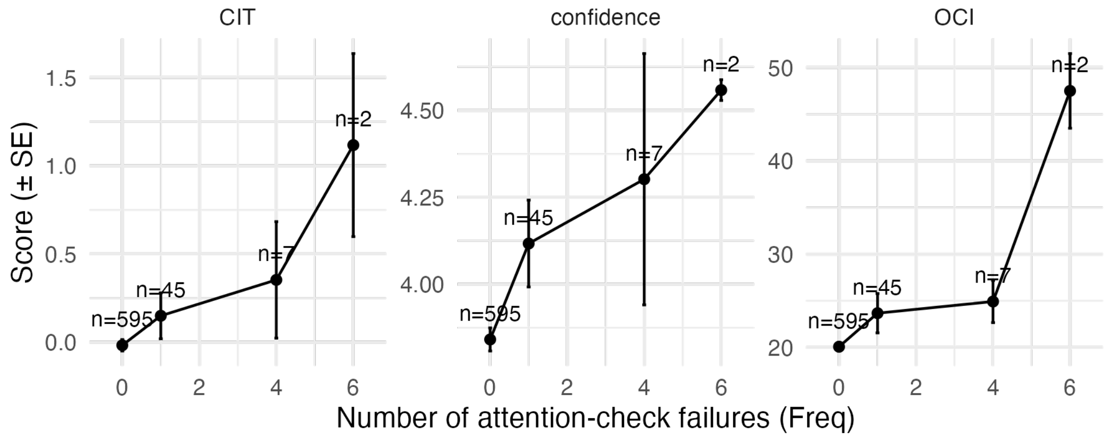

Figure A17 *- Inattentive participants in Fox et al., 2023 show elevated CIT, confidence, and OCI-R scores.*

In our experiment, which was designed to detect inattentive responding, we fitted a model that matched our data well and found that even with four infrequency items, only about 85% of inattentive participants could be identified (see Sarna et al. (35); Appendix, Model estimation of inattentive responding). This aligns well with previous work showing that statistically controlling for confounding variables using self-report measures is inherently difficult, and full control is rarely achieved (44). The reason is that self-report measures are noisy and provide only imperfect estimates of the underlying latent construct that they are intended to measure: a fact that is true of both infrequency items as a measure of inattentive responding and of neutral items as an estimate of acquiescence.

Beyond being difficult to fully control, both acquiescence and inattentive responding exert substantial influence on the variables of interest. In our data, the correlation between OCI-R and confidence after controlling for inattention and acquiescence, is $r = .11$, 95% CI $[-.05, .27]$, similar to the value reported by Gillan et al. (36) (n = 645), whereas the correlation between acquiescence and confidence is $r = .30$, 95% CI $[.17, .43]$. A model predicting confidence from acquiescence alone ($R^2 = .09$, 90% CI $[0.04, 0.17]$) accounts for roughly seven times more variance in confidence than a model with the corrected OCI-R scores ($R^2 = .01$, 90% CI $[0.00, 0.06]$). A model that includes both our inattention and acquiescence measures as regressors predicting confidence explains $R^2 = .14$, 90% CI $[0.06, 0.21]$ of the variance. This is roughly eleven times the variance explained by the corrected OCI-R in our data.

Admittedly, these comparisons are not straightforward, as the variance explained by the OCI-R is calculated after excluding inattentive participants, whereas the variance explained by surface-level questionnaire-filling behaviours is calculated with inattentive participants included (as the effects of inattention are of interest). And still, according to these estimates, to achieve 80% power to detect the correlation between OCI-R and confidence (controlling for inattention and acquiescence), one needs a sample of more than 600 participants, but to detect the correlation between confidence and acquiescence with the same power, a sample of 82 participants is sufficient, and to find an effect of inattentive responding in a between-group comparison, it is sufficient to have 42 participants per group. The relative magnitude of these required sample sizes is informative: it means that one needs to be very certain that all effects of acquiescence and inattentive responding are controlled for before concluding that a significant correlation between OCI-R and confidence is driven by more than undetected surface-level questionnaire-filling behaviours.

Taken together, given that the confounds we identified have effect sizes much larger than the effect of interest, and that full control for these confounds is not feasible, even a significant CIT–confidence correlation cannot rule out the possibility that these relationships are fully accounted for by the confounds. As Westfall & Yarkoni (44) note “the larger the influence of the confounding covariate, the more variance can be misattributed to the predictor of interest, leading to an increase in Type I error.”

3. Low statistical power

Gillan et al. (36) suggested that our sample size (n = 190) was insufficient to detect a significant compulsivity–confidence correlation. First, we wish to clarify that our intention was *not* to interpret our null finding as evidence of absence of a compulsivity-confidence correlation. Our main focus (as pre-registered: 10.17605/OSF.IO/JDQUY) was to test the magnitude of effects from surface-level questionnaire filling behaviours. This was quantified as correlations between questionnaire scores, mean confidence ratings, and our proxies of acquiescence and inattention, and as the relative drop in the OCI-R-confidence correlation when controlling for these confounds, not the final size of the correlation, as we explain in the Results section:

> *Readers should not over-interpret the nonsignificance of the final correlation, but instead focus on the magnitude of the drop in the correlation coefficient (from .27 to .11), which, as we show in the Appendix, was highly significant, and cannot be explained by the reduction in the sample size (excluding inattentive participants) nor by the correction procedure (regressing out the mean response to neutral items).*

Our sample was highly sensitive and well-powered to detect these huge effects. As we note in the previous section, even if the OCI-R-confidence correlation had remained significant after controlling for inattention and acquiescence, given the very modest (r^2=0.01) magnitude of the effect of interest relative to the large effect sizes of surface-level questionnaire-filling behaviour, and given that our measures of these confounds, advanced as they are, are never perfect, this would not count as firm evidence for a true correlation between confidence and obsessive compulsive tendencies.

We wish to end this section by noting the broader implication of increasing sample size when a confounding variable is measured unreliably. As Westfall and Yarkoni (44) clearly explain:

> *… as sample size increases, error rates also increase. It is worth reflecting on this result, because it contravenes the received wisdom that larger samples mitigate most common statistical problems (e.g., as n grows, power to reject the null increases, parameter estimates become more precise, etc.). Indeed, we find that for studies involving thousands of participants and non-negligible indirect effects, rejection of the null hypothesis is a near certainty even when the null is in fact true […] the reason for this behavior becomes clear: as samples grow, power to detect any reliable association between the predictors and the outcome necessarily increases. This remains true even*

> *when measurement unreliability causes the model to confuse a common effect of two or more predictors with a unique effect of one predictor—as n grows, the model more confidently concludes that there is a reliable association between the predictor of interest and the outcome.*

4. The use of single questionnaires instead of CIT and AD dimensions

Gillan et al. (36) argue that our findings based on the OCI-R cannot be generalized to the CIT dimension, as these capture different constructs. They rightly point out that the OCI-R reflects clinical OCD symptoms, while CIT represents a broader, transdiagnostic factor that includes compulsivity, among other related traits. Accordingly, they suggest that our conclusions rely on a false equivalence between these measures.

We therefore wish to make clear that it is not our claim that the OCI-R is equivalent to the CIT factor or that the SDS is equivalent to the AD dimension. Rather, our central point concerns the *ubiquity of response biases* in self-report measures commonly used in computational psychiatry and their effects of correlations with decision confidence, irrespective of their theoretical labeling. The psychometric features that render the OCI-R particularly vulnerable to response biases (its high positive skew and the absence of reversed-coded items) are shared by the items that constitute the CIT dimension. Because CIT is computed from multiple highly skewed items, and because it systematically undoes the semantic reversal of reversed-coded items by assigning them a negative weight, the mechanisms we identify are expected to operate similarly, if not more strongly. Consequently, the extent to which the OCI-R and CIT reflect identical clinical constructs is independent of our methodological concern: both are susceptible to the same forms of response bias.

As an argument for the robustness of the CIT-confidence correlation, Gillan et al. (36) emphasize that it has been repeatedly replicated across multiple datasets and studies. Yet replicating an effect across studies that share similar methodological vulnerabilities does not establish its validity as long as the same confounds are consistently operating. As such, repeated statistical significance cannot rule out the possibility that the CIT–confidence association, as currently observed, remains partly or wholly driven by residual response bias rather than genuine metacognitive effects. Unless demonstrated otherwise, we believe that the parsimonious assumption is that the same vulnerabilities apply across these instruments.

5. Skew is a feature, and also a bug

Gillan et al. (36) acknowledge that surface-level properties such as skewness and item coding are important and understudied, but argue that our assertions about their contribution to the transdiagnostic dimensions are not supported by data. Their claim is that skewness primarily reflects the underlying prevalence of symptoms: some mental health problems are common (e.g., depression and anxiety), while others are rare (e.g., compulsivity). Therefore, the association between skewness and CIT weights is expected.

We agree that skewness is not randomly assigned and that some symptoms are indeed rarer than others. In this sense, skewness is a feature of certain psychopathological constructs. However, skewness also has a bug component: rare response patterns are particularly vulnerable to inflation by inattentive responding.

Indeed, although skewness is a post-hoc measure and not designed to assess inattention, it aligns closely with inattention in practice. To show this, we reanalyzed data from Katyal et al. (2), which include a reliable inattention index based on infrequency items. For each psychiatric questionnaire item, we computed its sensitivity to inattention by comparing the difference in item endorsement between inattentive and attentive respondents (using Cohen's d to obtain a standardized effect size). If skewness is a sensitive index of inattention, items with higher skewness should show larger inattentive–attentive differences. This is exactly what we found: inattention sensitivity was strongly associated with item-level skewness ($r_s = .60$, $p < .001$; Figure A18, right panel). We also tested the correlation between CIT and AD factor weights and inattention sensitivity, and found the same patterns obtained for skewness: inattention sensitivity positively correlated with CIT weights ($r_s = .77$, $p < .001$; Figure A18, left panel), and a negatively correlated with AD weights ($r_s = -.56$, $p < .001$; Figure A18, middle panel). In other words, items that received higher ratings from inattentive participants loaded high on CIT and low on AD, and the opposite was true for items that received lower ratings from inattentive participants. This validates our use of skewness as a proxy for inattention sensitivity, and reaffirms that the factor structure strongly aligns with inattentive responding.

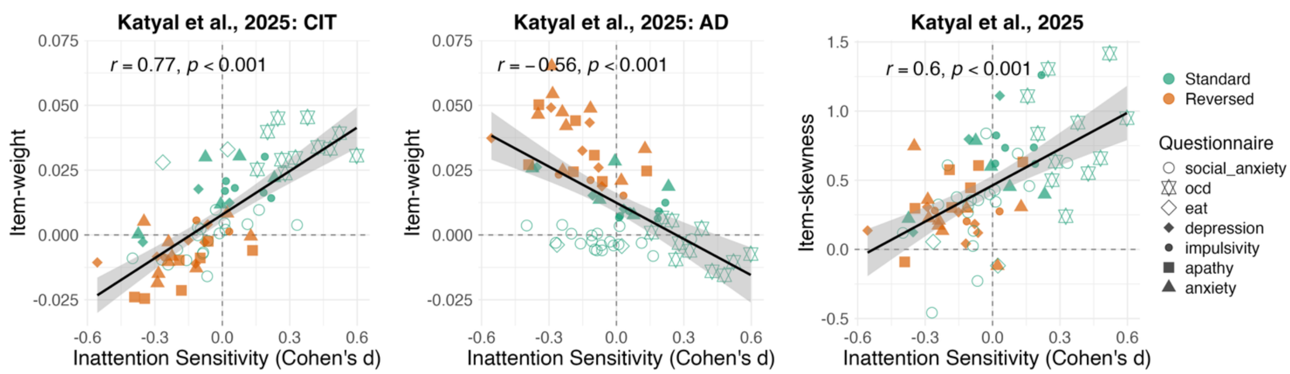

Figure A18 *- Item sensitivity to inattention is correlated with both CIT/AD item weights and item-level skewness in Katyal et al. (2025).* Each point represents a questionnaire item, with point shapes indicating the questionnaire and color coding for standard versus reversed items. The x-axis shows each item's sensitivity to inattentive responding (Cohen's d, comparing inattentive vs. attentive participants). Left panel: the y-axis shows the CIT item-weight. Middle panel: the y-axis shows the AD item-weight. Right panel: the y-axis shows item-level skewness. The solid black line represents the linear regression fit, and the shaded region reflects the standard error of the fit. The reported correlation is the Spearman correlation coefficient.

6. Additional clarifications

Gillan et al. (36) also noted two analysis errors found in earlier versions of our preprint. First, in the acquiescence correction procedure for the SDS questionnaire, the residualized scores were misaligned with participant IDs. Second, in the acquiescence correction procedure for both the OCI-R and the SDS, the model predicting item ratings from the mean response to neutral items included inattentive responders who should have been excluded prior to the correction. We are grateful to Gillan et al. (36) for identifying these errors. Both issues have been corrected in the current version of the preprint and, importantly, had no effect on the pattern of results or their interpretation. These corrections had only minor influence on some descriptive values, detailed below:

15. In the main text Results section, under "Inattentiveness and acquiescence are associated with shifts in confidence ratings," the correlation between the OCI-R (excluding inattentive responders and controlling for acquiescence) changed from $r = .09$, 95% CI $[-.08, .25]$, $t(142) = 1.03$, $p = .304$ to $r = .11$, 95% CI $[-.05, .27]$, $t(142) = 1.36$, $p = .176$. Additionally, the correlation between the SDS (excluding inattentive responders and controlling for acquiescence) changed from $r = -.13$, 95% CI $[-.28, .04]$, $t(142) = -1.52$, $p = .131$ to $r = -.13$, 95% CI $[-.29, .04]$, $t(142) = -1.55$, $p = .124$.

16. In the Appendix, under "A bootstrap permutation test for response style effects on OCI-confidence correlations," the small change in the OCI-R correlation mentioned above also altered the drop in the correlation coefficient after controlling for acquiescence from 0.19 to 0.17.

7. What we know, and what we still don't know

We believe our work convincingly demonstrates the ubiquity of response biases in online samples, their sizable effects on correlations between self-reported mental health and confidence ratings, and the difficulty to fully control for these effects. We also think these results cast many previous findings in a different light, raising questions about their interpretability. Importantly, *we do not claim that all correlations between self-reported mental health and confidence ratings are spurious, or are "explained away".* It is however the case that more work is needed to determine which findings are fully explained by these biases and which are only partially accounted for. For instance, when considering the diverging metacognitive patterns between individuals with high compulsive tendencies and those with clinical OCD (45), we acknowledge that while some important pieces of this puzzle were uncovered by our original paper, others remain unresolved. In particular, we still do not know why online studies do not show a negative correlation between OC tendencies and confidence, as would be expected from the centrality of self-doubt in OCD phenomenology. We note that this tension is not resolved by the difference between obsessive compulsive tendencies (as measured with the OCI-R questionnaire) and compulsivity (as reflected in the CIT dimension). In our study, we find a positive correlation between OCI-R and confidence, which does not become negative when removing inattentive participants and controlling for acquiescence. Additional possibilities that will need to be examined in future work concern the testing environment (online vs. in-person), the cognitive domain (memory vs. perception), and the relevance of these variables to OCD phenomenology (see also the Discussion in Sarna et al. 46). We leave it for future work to fully disentangle the OCD–compulsivity paradox.

Reproducibility

Data availability

Data is available on OSF at https://osf.io/75vc2/.

Code availability

Our fully reproducible analysis code is available on GitHub at https://github.com/self-model/reply-to-reply.

R package versions

```
##    effsize     ggpubr    moments    ggrepel  patchwork effectsize 
osfr 
##    "0.8.1"    "0.6.0"   "0.14.1"    "0.9.5"    "1.2.0"    "0.8.7" 
"0.2.9" 
##     papaja tinylabels  lubridate    forcats    stringr      purrr 
readr 
##    "0.1.2"    "0.2.4"    "1.9.3"    "1.0.0"    "1.5.1"    "1.0.2" 
"2.1.5" 
##      tidyr     tibble  tidyverse  gridExtra      dplyr    ggplot2 
##    "1.3.1"    "3.2.1"    "2.0.0"      "2.3"    "1.1.4"    "3.5.0"
```

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